# Solid State Physics — Week 4 Assignment Questions & Answers

## Question 31

A plane intercepts the crystal axes along 'a' at 0.25a and along 'b' at 0.50b and it is parallel to the third axis. Determine the Miller indices of the plane.

(a) (210)

(b) (310)

(c) (410)

(d) (110)

**Accepted answer: (a)**

## Question 32

Nickel has FCC structure with a lattice parameter of 3.52 Å. Find the interplanar spacing between the closest packed planes (111) in the metal.

(a) 1.05 Å

(b) 1.87 Å

(c) 2.03 Å

(d) 3.03 Å

**Accepted answer: (c)**

Question 33

Iron has BCC structure below 910°C and FCC structure above this temperature. Assume the same atomic radius in both structures. Find the percentage change in volume, when the metal undergoes the structural change at 910°C.

(a) *Volume is contract by* 8.14%

(b) *Volume is increases by* 10.24 %

(c) *Volume is contract by* 83.12 %

(d) *Volume is increases by* 23.14 %

**Accepted answer: (a)**

Question 34

Calculate the value of cation to anion radius ratio $\left(\frac{r}{R}\right)$ for CsCl structure in the limiting case when the anions are as close as possible. Where r is the radius of the cation and R is the radius of the anion. CsCl structure can be viewed as a BCC lattice with atoms of one type (anion) at the corners and the atoms of the other type (cation) at the body centers.

(a) 0.023

(b) 0.329

(c) 0.545

(d) 0.732

**Accepted answer: (d)**

Question 35

Calculate the density of CsCl given that radii of the $Cs^+$ ion = 1.65Å and for $Cl^-$ ion = 1.81Å. Atomic weights of Cs and Cl are 132.9 gm/mol and 35.45 gm/mol respectively. Consider one formula unit contain within the unit cell.

(a) $4.4 \times 10^3\ kg/m^3$

(b) $5.5 \times 10^3\ kg/m^3$

(c) $6.6 \times 10^3\ kg/m^3$

(d) $7.7\ kg/m^3$

**Accepted answer: (a)**

Question 36

Cesium metal (atomic weight = 130) has a cubic unit cell of side 6Å. If the density of Cs is $2\ gm/cm^3$, determine the type of its crystal structure.

(a) Simple cubic structure

(b) Face centred cubic structure

(c) Body centred cubic structure

(d) Hexagonal closed packed structure

**Accepted answer: (c)**

Question 37

If the lattice parameter for FCC crystals 2.14 Å, find out its value of atomic radius.

(a) 0.76 Å

(b) 0.85 Å

(c) 0.97 Å

(d) 1.23 Å

**Accepted answer: (a)**

Question 38

NaCl crystal has a cubic structure and its density is 2.17 $gm/cm^3$. Find out the distance between the two adjacent atoms of the same kind. Atomic weight of Na is 23 gm/mol and for Cl is 35.45 gm/mol. There are four formula units contain within the unit cell.

(a) 5.63 Å

(b) 5.87 Å

(c) 6.27 Å

(d) 6.57 Å

**Accepted answer: (a)**

Question 39

In a tetragonal crystal, the lattice parameters a = b = 2.42 Å and c = 1.74Å. Find out the interplanar spacing between consecutive (101) planes in the tetragonal crystal structure.

(a) 1.41 Å

(b) 1.52 Å

(c) 1.72 Å

(d) 1.93 Å

**Accepted answer: (a)**

Question 40

A solid with FCC-structure has a lattice constant 4 Å. Each atom in the solid has a mass of$4 \times 10^{-26}\ kg$. Calculate the number of conduction electrons in $1\ m^3$ volume of the solid, if each of the atoms of the solid contributes two electrons to its conduction band.

(a) $1.57 \times 10^{28}/m^3$

(b) $1.25 \times 10^{29}/m^3$

(c) $2.21 \times 10^{30}/m^3$

(d) $4.45 \times 10^{32}/m^3$

**Accepted answer: (b)**